MIT | Academy of Engineering

Department of Computer Engineering

Presentation for TY Major Project , 2025-26

HAWK SIGHT AI: Ai Stock Market Prediction Model

Students Name:

Mandar Ghadge (202301100024)

Darshan Bhabad (202301040169)

Nikhil Bodre (202301100008)

Atharav Barapatre (202301040077)

Guide:

Mr. Chaitanya Patil

Index

- Introduction
- Objectives
- Literature Survey
- Project Design & Diagrams
- System Architecture
- Methodology & Algorithm
- Conclusion & Future Scope

Introduction

The Problem:

- Financial markets are characterized by non-linear volatility and massive data generation.
- Traditional static analysis tools (charts/indicators) often lag behind real-time market shifts.
- Manual trading lacks the speed to capitalize on micro-trends.

The Solution (Hawk Sight AI):

- **Hawk Sight AI** is a sophisticated web-based application designed to bridge the gap between raw financial data and actionable intelligence.
- It leverages **Deep Learning (LSTM Neural Networks)** to analyze historical trends (Price, RSI, EMA) and forecast future price movements.
- Unlike static models, this system implements a **"Continuous Learning"** loop, retraining itself on the latest data before every single prediction.

Objectives

1. **Predictive Accuracy:** To develop an LSTM-based deep learning model capable of minimizing Root Mean Square Error (RMSE) in stock price prediction.
2. **Real-Time Processing:** To implement a low-latency data pipeline using Finnhub APIs and Flask, ensuring market data is live.
3. **Dynamic Adaptation:** To engineer a self-training mechanism where the model fine-tunes on the last 150 days of data instantly upon user request.
4. **Enhanced UX/UI:** To design an immersive "Glassmorphism" interface that visualizes complex AI confidence scores and trend indicators intuitively.

Literature Survey

Sr. No.	Author (Year)	Paper Title	Key Methodology / Findings
1	**Mishra et al. (2021)**	Stock Price Prediction Using LSTM & RNN	Found LSTM superior to standard RNNs for long-term dependencies due to the vanishing gradient problem.
2	**Zhang & Liu (2020)**	Hybrid CNN and Deep Learning for Stock Prediction	Combined CNN for feature extraction with LSTM for temporal patterns; improved accuracy by 12%.
3	**Mondal & Mandal (2019)**	Study of RNN with Limited Memory	Highlighted the limitations of standard RNNs in remembering long sequences, motivating the use of LSTM.
4	**Bao et al. (2019)**	Wavelet-based Deep Neural Network	Used wavelet transforms to denoise data before feeding it into deep networks.

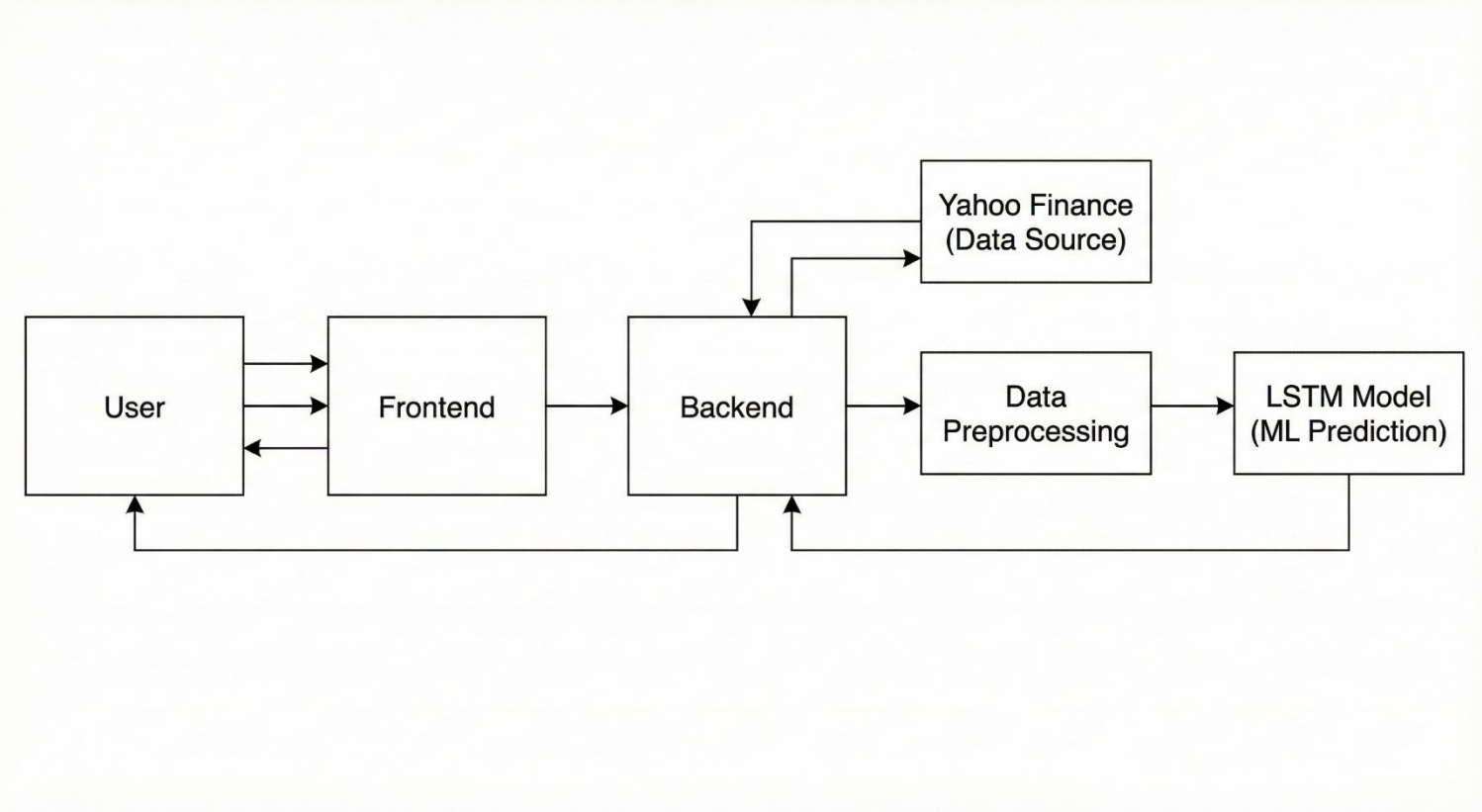
Literature Survey

Sr. No.	Author (Year)	Paper Title	Key Methodology / Findings
5	**Hu et al. (2020)**	LSTM-RNN Model for Trend Forecasting	Focused specifically on "Trend" (Direction) prediction rather than exact price values.
6	**Sardar et al. (2021)**	Hybrid LSTM and Extreme Learning Machines (ELM)	Proposed ELM for faster training times, though with slightly lower accuracy than pure LSTM.
7	**Chen & He (2018)**	Stock Prediction Using CNN in Chinese Markets	Demonstrated that technical indicators (RSI, MACD) significantly improve model performance when added as features.
8	**Roondiwala et al. (2017)**	Predicting NIFTY prices using LSTM	Early foundational paper proving LSTM can map NIFTY 50 volatility effectively.

Project Design & Diagrams

Block Diagram

•**Data Acquisition:** Raw data is fetched from Yahoo Finance (Historical) and Finnhub (Live).
•**Preprocessing:** Data is cleaned, and indicators (RSI, EMA-50) are calculated. MinMaxScaler normalizes values between 0 and 1.
•**Processing Unit:** The LSTM Model (2 Layers + Dropout) processes the sequence.
•**Output:** The model outputs a scalar value (Next Day Price), which is inverse-scaled to get the actual dollar amount.

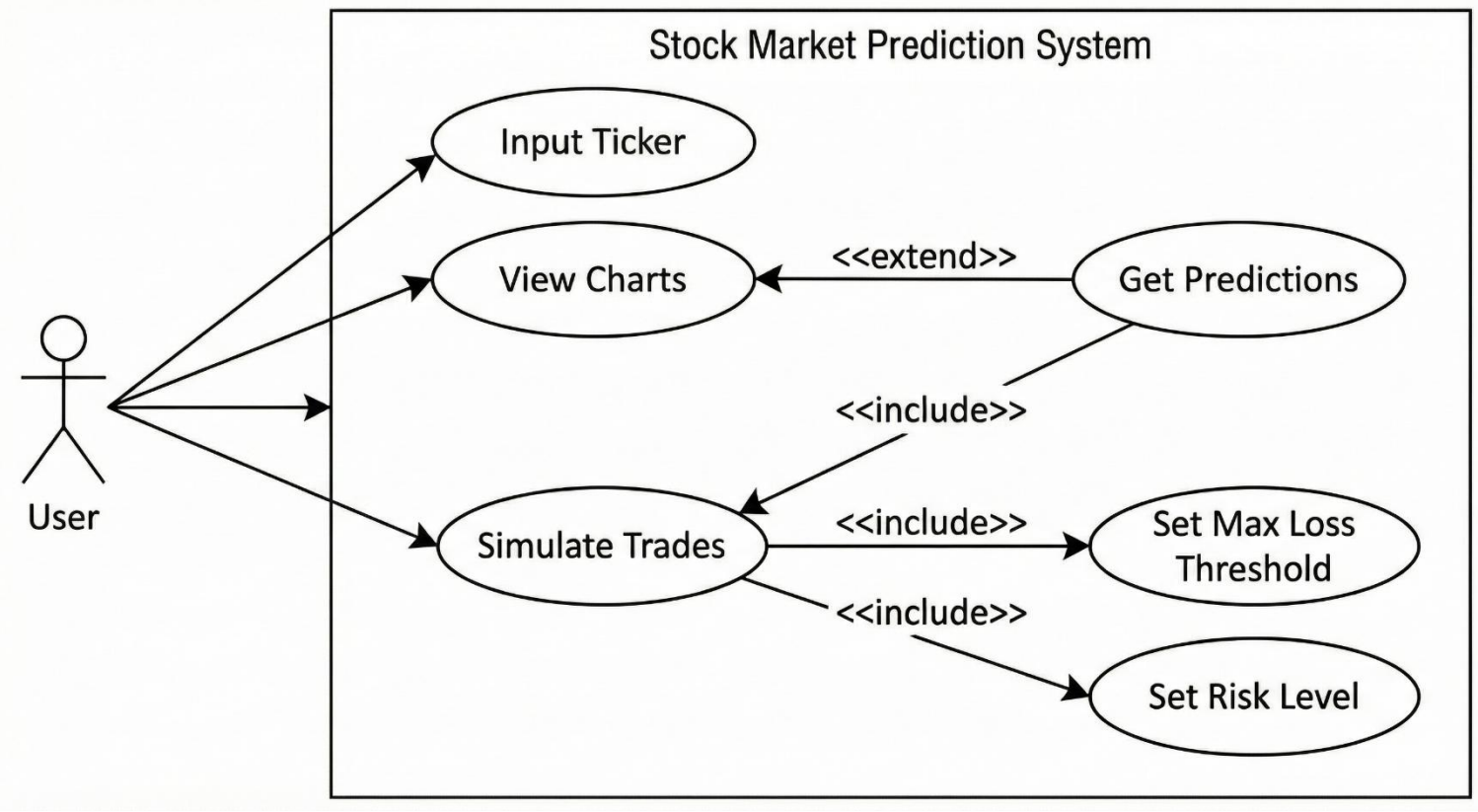


Project Design & Diagrams

Usecase Diagram

Relevance: Defines the interactions between the users and the system components.

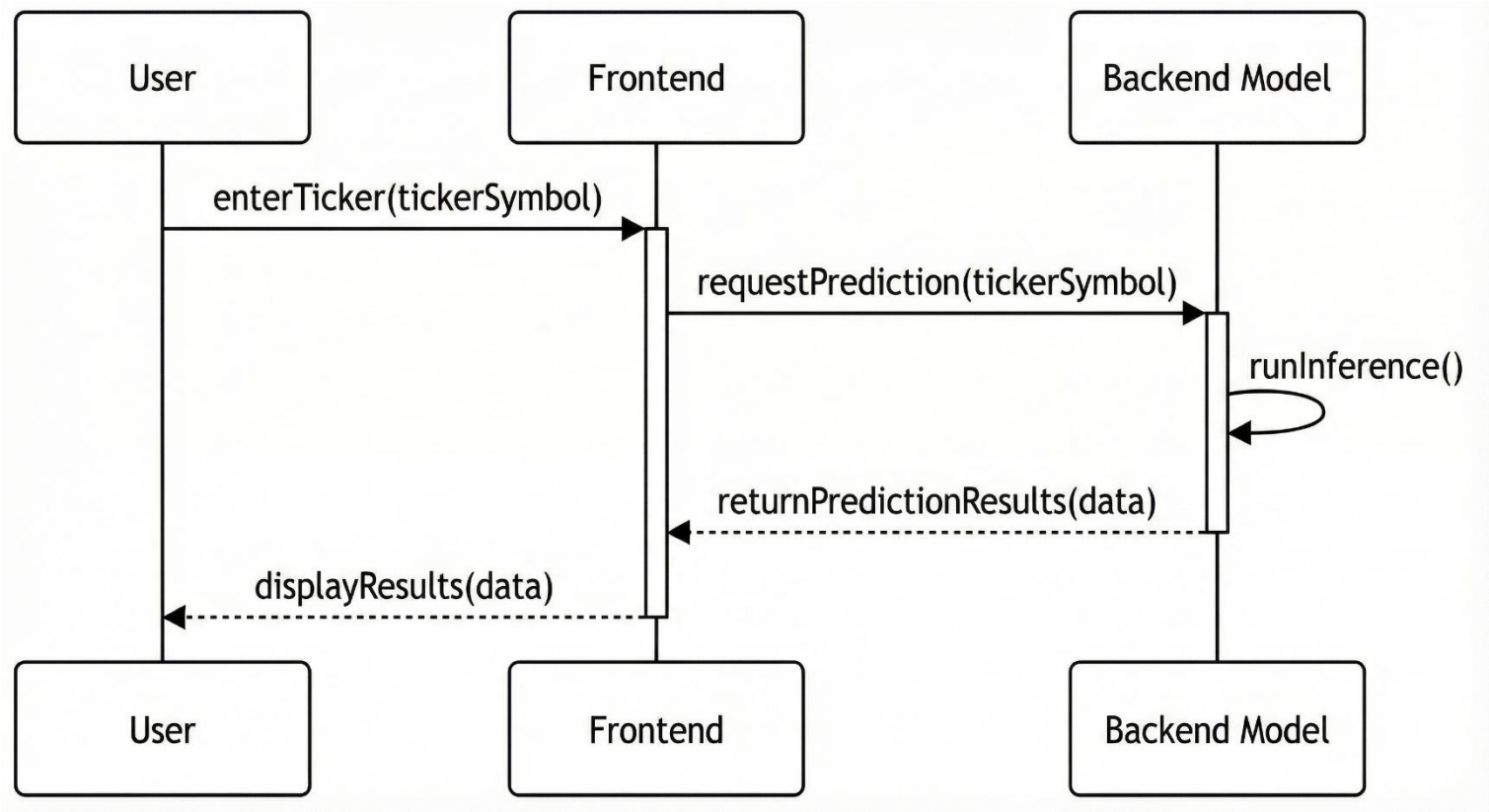
- **Actor: User (Trader)**
  - Select Stock Ticker.
  - View Real-time Dashboard.
  - Trigger AI Prediction.
- **Actor: System (AI Engine)**
  - Fetch Historical Data.
  - Calculate Technical Indicators.
  - **Self-Train (Fine-tune Model).**
  - Generate Buy/Sell Signal.



Project Design & Diagrams

Sequence Diagram

Relevance: Details the chronological order of operations for a single prediction request.
Step 1: UI sends POST /predict {symbol: 'AAPL'} to Flask Server.
Step 2: Server initializes StockPredictor class.
Step 3: Predictor fetches last 2 years of data from Yahoo Finance.
Step 4: Predictor calculates RSI & EMA, then fine-tunes the model on the last 150 days.
Step 5: Model predicts next step; Server calculates RMSE/R2.
Step 6: Server returns JSON response; UI updates the "Projected Price".



System Architecture

1. **Model (The Brain - Python/Keras):**

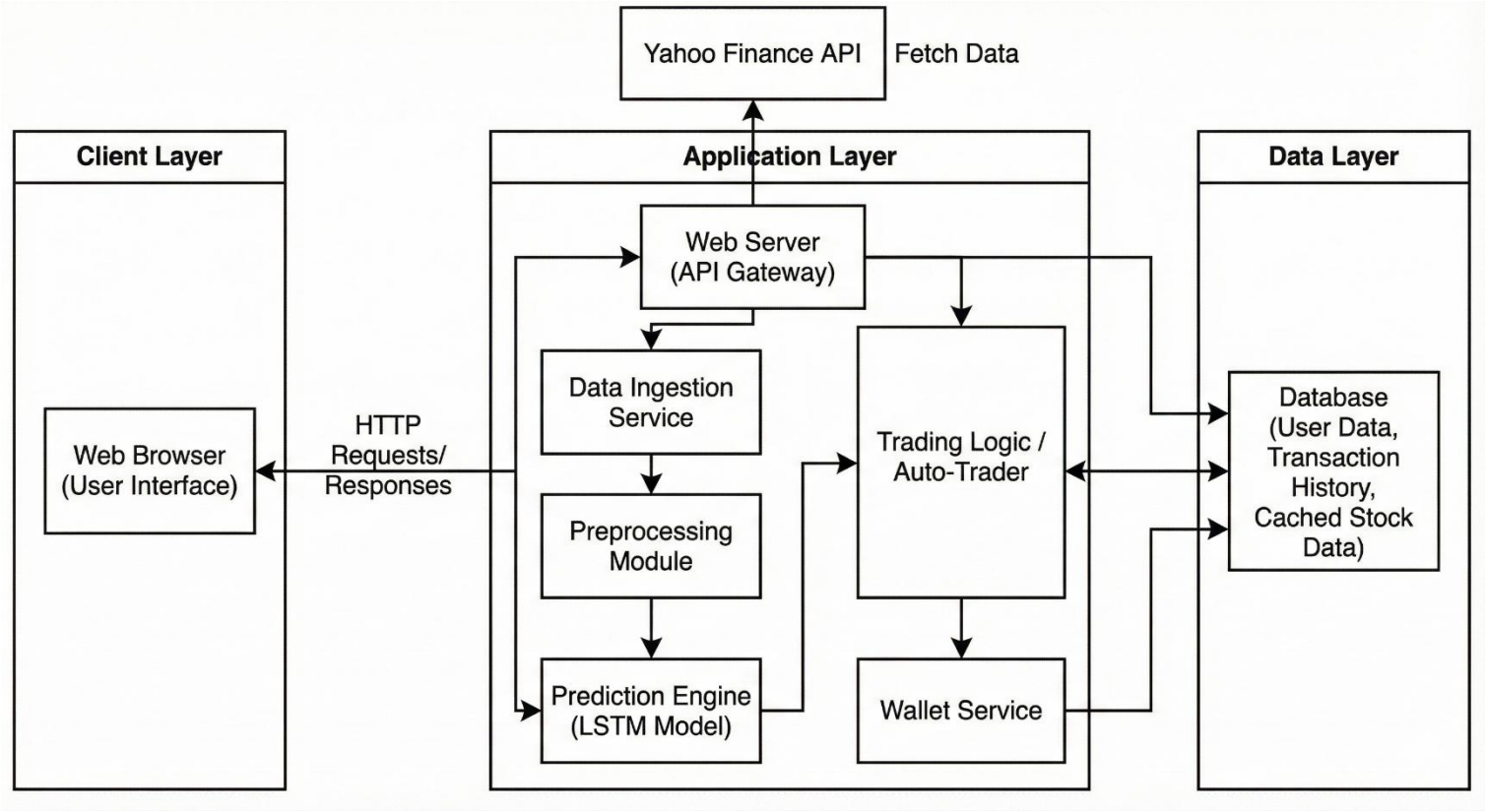
•**File:** stock_model.py
•Contains the LSTM Neural Network logic.
•Handles feature engineering (RSI, EMA) and data normalization.
•Manages the "Self-Training" loop.

2. View (The Face - HTML/Tailwind):

•**File:** index.html
•A responsive web interface built with Tailwind CSS.
•Uses Glassmorphism design principles for a modern look.
•Visualizes data via dynamic DOM manipulation.

3. Controller (The Bridge - Flask):

•**File:** server.py
•Exposes a REST API (/predict).
•Receives user inputs, triggers the Model, and formats the response for the View.



Methodology & Algorithm

The Algorithm: Long Short-Term Memory (LSTM)

•**Why LSTM?** Standard RNNs suffer from the "vanishing gradient" problem, making them forget early data in long sequences. LSTMs use "gates" to retain relevant long-term history.

•**Architecture Specifics:**

- **Input Layer:** Sequence of 60 Days x 3 Features (Close, RSI, EMA).
- **Hidden Layer 1:** LSTM (64 Units) + Dropout (0.2) to prevent overfitting.
- **Hidden Layer 2:** LSTM (64 Units).
- **Output Layer:** Dense (1 Unit) for price regression.

•**The "Self-Learning" Innovation:**

- Unlike static models, Hawk Sight checks if a pre-trained model exists.
- If yes, it loads it and **immediately trains** on the most recent 100 days of data before making a prediction.
- This allows the AI to adapt to breaking news or recent volatility instantly.

Conclusion & Future Scope

Conclusion: Hawk Sight AI successfully demonstrates that integrating deep learning with real-time financial data can yield actionable insights. By employing a hybrid "Pre-trained + Self-learning" approach, the system overcomes the issue of static models becoming obsolete in volatile markets.

Future Scope:

Sentiment Analysis: Integrating NLP to analyze news headlines (Twitter/X API) to adjust predictions based on public sentiment.

Multi-Step Forecasting: Extending the model to predict the next 7 days instead of just 24 hours.

Live Deployment: Migrating the Flask server to a cloud provider (AWS/Heroku) for global access.

Thank You!!

Any questions.